# Nidhish Kumar Singh

Full-Stack Software Engineer | Java · Spring Boot · React.js · AI/LLM Systems

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## PROFESSIONAL SUMMARY

Software Engineer with nearly 3 years at Cognizant (client: LexisNexis), specialising in backend development with **Java / Spring Boot** and exposure to frontend development in React.js. Experienced in designing REST APIs and microservices, securing applications with Spring Security, and working with SQL Server and PostgreSQL. Also hands-on with LLM integration, with a solid grasp of design patterns, system design and CI/CD.

## PROFESSIONAL EXPERIENCE

**Cognizant Technology Solutions | Client: LexisNexis** Dec 2023 - Present

*Software Engineer, Backend* *Bengaluru, India*

- **Architected a multi-agent AI pipeline** (Generator + Verifier agents) that auto-generates document-parsing configurations, replacing hand-written extraction code across **250+ legal data sources**, cutting new-source delivery from **4 weeks to under 2 days** at **90%+ accuracy**.
- Built a centralised **AI gateway microservice** routing across multiple LLM providers (OpenAI, Anthropic, DeepSeek via Azure AI Foundry) with **priority-based fallback, circuit-breaker health checks** and provider-agnostic **prompt caching that cut cached-input token cost by ~90%**, used by **4+ consumer applications**.
- **Upgraded a legacy Spring Boot application from Java 7 to Java 17** during a scaling initiative, migrating **12 modules** to Streams, lambdas and Optional, cutting boilerplate by roughly a third and eliminating a recurring class of null-pointer defects.
- **Implemented OAuth2 single sign-on** for a Spring Boot application with **stateless JWT authentication**, enforcing **role-based access** (USER / ADMIN) on every endpoint that handles personal user data.
- Designed a **durable job queue** on SQL Server with a dedicated worker service for background AI config generation, with safe job claiming across instances, automatic recovery of abandoned jobs, and quality and token tracking per request.
- Developed a **React + Spring Boot dashboard** integrating Azure DevOps APIs to track real-time ticket throughput and SLA compliance, reducing reporting overhead by **40%**.
- Built a **database automation platform** monitoring **250+ SQL Server instances** with automated health checks and anomaly detection, reducing manual DBA work by **40%**; wrote automated API tests and managed **Azure DevOps CI/CD** deployments.

## TECHNICAL SKILLS

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|---|---|
| **Languages:** | Java, C#, SQL, Python, JavaScript |
| **Backend:** | Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices |
| **AI & LLM:** | RAG Pipelines, Vector Search (pgvector), LangChain4j, Prompt Engineering, Multi-Provider LLM Orchestration, Agentic Workflows, Azure AI Foundry, AI-Assisted Development (Claude Code, Cursor) |
| **Frontend:** | React.js, JavaScript, HTML, CSS |
| **Databases:** | PostgreSQL, SQL Server |
| **Tools & DevOps:** | Docker, Git, Azure DevOps CI/CD, Postman |
| **Fundamentals:** | Data Structures & Algorithms, OOP, Multithreading, Design Patterns, System Design |

## PROJECTS

**ResuMatch: AI-Powered Job Recommendation Engine |** github.com/nidkdevtech/resumatch

*Tech: Java, Spring Boot, LangChain4j, PostgreSQL, pgvector, Ollama, Groq, React.js, Docker*

- Built an end-to-end **RAG pipeline** with LangChain4j and pgvector for semantic resume-to-job matching across **1,100+ real job listings**, replacing keyword search with vector similarity.
- Designed a **provider-agnostic embedding layer** supporting Ollama and OpenAI through configuration-driven switching using the Strategy pattern, so providers swap with zero code changes.
- Implemented **hybrid search** combining pgvector cosine similarity with SQL filters and window functions for chunk deduplication, ensuring each job appears exactly once in results.
- Integrated **Groq LLaMA 3** for AI match explanations with batched prompt processing (5 jobs/call) to optimise context-window usage.

**Smart Expense Tracker |** github.com/nidkdevtech/SmartExpenseTracker

*Tech: Java, Spring Boot, Spring Security, PostgreSQL, Stripe API, Groq, React.js*

- Implemented **JWT authentication** with Spring Security and role-based access control (USER/ADMIN), securing all API endpoints with fine-grained authorisation.
- Integrated the **Stripe** payment gateway for subscription management, handling webhook events, tracking payment history and gating premium features on subscription status.
- Integrated Groq LLaMA 3 via an OpenAI-compatible API to generate personalised monthly financial insights from categorised expense patterns.

## ACHIEVEMENTS

- **Anthropic Certifications:** Claude Certified Architect - Foundations and Claude Code in Action (Jul 2026). Production LLM application architecture and AI-assisted development with coding agents.
- **Project Key Contributor** (Cognizant, Sep 2025): recognised for designing and leading development of the AI-powered config generation system.

## EDUCATION

**GL Bajaj Institute of Technology and Management** Jul 2019 - Jun 2023

*Bachelor of Technology (B.Tech), Electronics and Communication Engineering* *Greater Noida, India*